The documentation and process conversion measures necessary to comply with this revision shall be completed by 8 May 1995

INCH-POUND

MIL-S-19500/177E
8 February 1995
SUPERSEDING
MIL-S-19500/177D
30 December 1991

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER
TYPES 2N1131, 2N1131L, 2N1132, 2N1132L, JAN AND JANTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNP silicon, low-power transistors. Two levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (TO-5, TO-39).

1.3 Maximum ratings.

P_T 1/ T_C = +25°C	P_T 2/ T_A = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_{OP} and T_J
W	W	V dc	V dc	V dc	mA dc	°C
2.0	0.6	50	40	5.0	600	-65 to +200

1/ Derate linearly 11.4 mW/°C for $T_C \geq$ +25°C.
2/ Derate linearly 3.4 mW/°C for $T_A \geq$ +25°C.

1.4 Primary electrical characteristics.

	h_{FE1} 1/		h_{fe1}		$V_{BE(SAT)}$ 1/		$V_{CE(SAT)}$ 1/		C_{obo}		$\|h_{fe}\|$	
	V_{CE} = 10 V dc, I_C = 150 mA dc		V_{CE} = 5.0 V dc, I_C = 1.0 mA dc, f = 1 kHz		I_C = 150 mA dc, I_B = 15 mA dc		I_C = 150 mA dc, I_B = 15 mA dc		f = 1 MHz, V_{CB} = 10 V dc, I_E = 0		f = 20 MHz, I_C = 50 mA dc, V_{CE} = 10 V dc	
	Min	Max	Min	Max	V dc Min	V dc Max	V dc Min	V dc Max	pF Min	pF Max	Min	Max
2N1131,L	20	45	15	50		1.3		1.5		45	2.5	20
2N1132,L	30	90	30	90		1.3		1.5		45	3.0	20

1/ Pulsed (see 4.4.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765 by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500, and as follows:

R_{BE} . . . . . External resistance, base to emitter.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figure 1 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be gold, tin-plated or solder. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2.1 Thermal resistance. Thermal resistance testing shall be performed in accordance with MIL-STD-750, method 3151 for qualification only.

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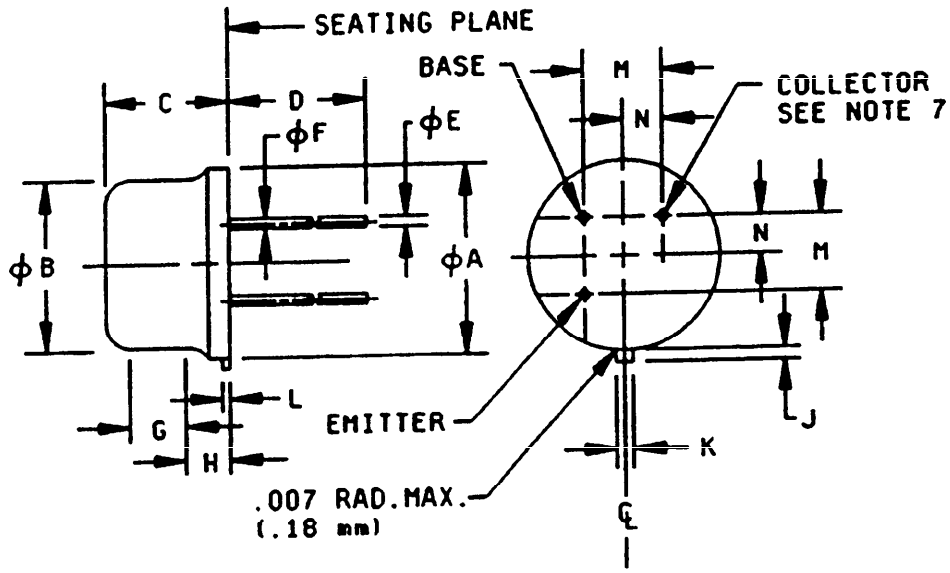


FIGURE 1. Physical dimensions.

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Ltr	Dimensions				Notes	Ltr	Dimensions				Notes
	Inches		Millimeters				Inches		Millimeters		
	Min	Max	Min	Max			Min	Max	Min	Max	
ϕA	.335	.370	8.51	9.40		H					6
ϕB	.305	.335	7.75	8.51	5	J	.029	.045	0.74	1.14	9
C	.240	.260	6.10	6.60		K	.028	.034	0.71	0.86	
D	See notes				10,11,12	L	.009	.125	0.23	3.18	
ϕE	.016	.021	.41	.53	3	M	.1414 Nom		3.59 Nom		7
ϕF	.016	.019	.41	.48	4	N	.0707 Nom		1.80 Nom		7
G	.100		2.54								

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Measured in the zone beyond .250 inch (6.35 mm) from the seating plane.
4. Measured in the zone .050 inch (1.27 mm) and .250 inch (6.35 mm) from the seating plane.
5. Variations on dimension ϕB in zone G shall not exceed .010 inch (0.25 mm). This zone is controlled for automatic handling.
6. Outline in this zone is not controlled.
7. When measured in gauging plane .054 +.001, -.000 inch (1.37 +0.03, -0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 inch (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 shows the preferred method of measurement.
8. The collector shall be electrically connected to the case.
9. Measured from the maximum diameter of the actual device.
10. All 3 leads (see 3.3.1).
11. For transistor types 2N1131 and 2N1132, dimension D shall be 0.500 inch (12.70 mm) minimum and 0.750 inch (19.05 mm) maximum.
12. For transistor types 2N1131L and 2N1132L, dimension D shall be 1.500 inches (38.10 mm) minimum and 1.750 inches (44.45 mm) maximum.

FIGURE 1. Physical dimensions - Continued.

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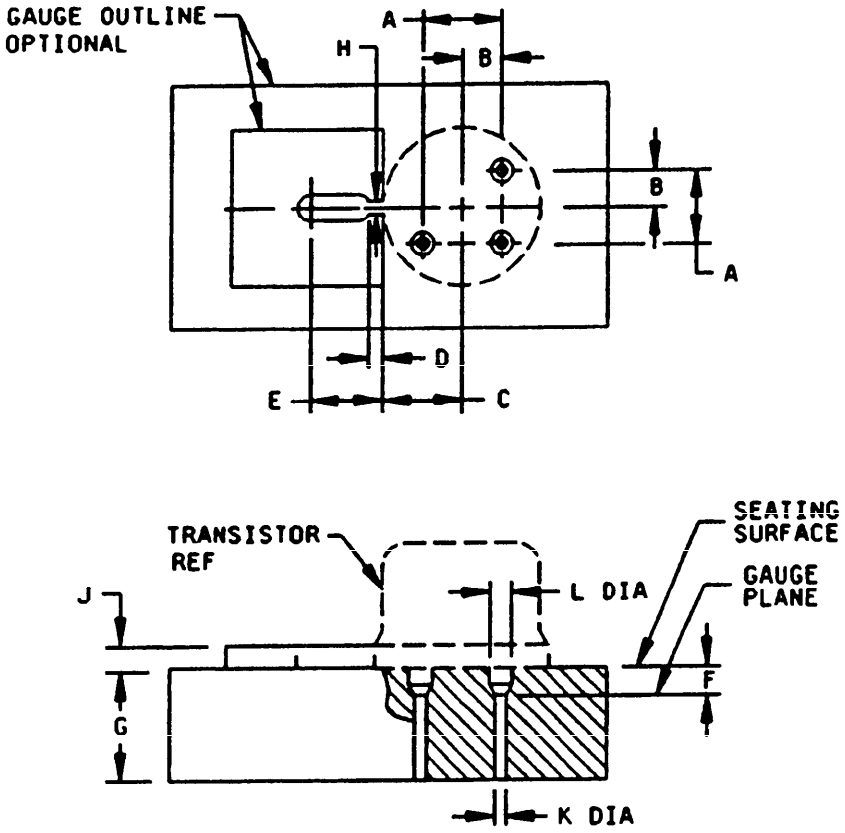


FIGURE 2. Gauge for lead and tab location.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.1409	.1419	3.578	3.604	
B	.0702	.0712	1.783	1.808	
C	.182	.199	4.62	5.05	
D	.009	.011	.23	.28	
E	.125 Nom		3.18 Nom		
F	.054	.055	1.37	1.40	
G	.372	.378	9.45	9.60	
H	.0350	.0355	.889	.902	
J	.150 Nom		3.81 Nom		
K	.0325	.0335	0.825	0.850	
L	.0595	.0605	1.51	1.54	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The following gauging procedure shall be used: The use of a pin straightener prior to insertion in the gauge is permissible. The device being measured shall be inserted until its seating plane is .125 ±.010 inch (3.18 ±0.25 mm) from the seating surface of the gauge. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application. A force of 8 ounces ±.50 ounce shall then be applied parallel and symmetrical to the advice's cylindrical axis. When examined visually after the force application (the force need not be removed), the seating plane of the device shall be seated against the gauge.
4. The location of the tab locator, within the limits of dimension C, will be determined by the tab and flange dimension of the device being checked.

FIGURE 2. Gauge for lead and tab location - Continued.

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4.3 Screening (JANTX level only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurement shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screening (see table II of MIL-S-19500)	Measurements JANTX level only
9	Not applicable
11	I_{CBO} and h_{FE2}
12	Not applicable
13	ΔI_{CBO} = 100 percent of initial value or 10 µA dc whichever is greater; Δh_{FE2} = ±20 percent of initial value; subgroup 2 of table I herein.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX) of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Conditions
B3	1027	T_A = room ambient as defined in the general requirements of MIL-STD-750, V_{CB} = 30 V dc; I_C = 20 mA dc.
B3	2037	Test condition A.
B5	3161	For qualification only.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Conditions
C2	2036	Test condition E.
C6	1026	T_A = room ambient as defined in the general requirements of MIL-STD-750, V_{CB} = 30 V dc; I_C = 20 mA dc.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector - emitter breakdown voltage	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	40		V dc
Collector - base breakdown voltage	3001	Bias condition D; I_C = 10 mA dc	$V_{(BR)CBO}$	50		V dc
Collector - base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO1}		10	µA dc
Collector - emitter cutoff current	3041	Bias condition B; V_{CE} = 50 V dc; pulsed (see 4.5.1); $R_{BE} \leq$ 10 ohms	I_{CER}		10	mA dc
Collector - base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO2}		1.0	µA dc
Emitter - base cutoff current	3061	Bias condition D; V_{EB} = 5.0 V dc	I_{EBO}		100	µA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; Pulsed (see 4.5.1)	h_{FE1}			
2N1131, L 2N1132, L				20 30	45 90	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 5.0 mA dc; pulsed (see 4.5.1)	h_{FE2}			
2N1131, L 2N1132, L				15 25		
Collector - emitter saturated voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(SAT)}$		1.5	V dc
Base - emitter saturated voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(SAT)}$		1.3	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector - base cutoff current	3036	Bias condition D; V_{CE} = 30 V dc	I_{CBO3}		100	µA dc
Low-temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE3}			
2N1131, L				10		
2N1132, L				15		
Subgroup 4						
Small-signal short circuit forward-current transfer ratio	3206	V_{CE} = 5.0 V dc; I_C = 1.0 mA dc; f = 1 kHz	h_{fe1}			
2N1131, L				15	50	
2N1132, L				30	90	
Small-signal short circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 5.0 mA dc; f = 1 kHz	h_{fe2}			
2N1131, L				20		
2N1132, L				30		
Small-signal open-circuit output admittance	3216	V_{CE} = 5.0 V dc; I_C = 1.0 mA dc	h_{ob1}	0.0	1.0	
Small-signal open-circuit output admittance	3216	V_{CE} = 10 V dc; I_C = 5.0 mA dc	h_{ob2}	0.0	5.0	

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Small-signal short-circuit input impedance	3201	V_{CE} = 5.0 V dc; I_C = 1.0 mA dc	h_{ib1}	25	35	
Small-signal short-circuit input impedance	3201	V_{CE} = 10 V dc; I_C = 5.0 mA dc	h_{ib2}		10	
Small-signal open-circuit reverse-voltage transfer ratio	3211	V_{CE} = 5.0 V dc; I_C = 1.0 mA dc	h_{rb1}	0.0	$8x10^{-4}$	
Small-signal open-circuit reverse-voltage transfer ratio	3211	V_{CE} = 10 V dc; I_C = 5.0 mA dc	h_{rb2}	0.0	$8x10^{-4}$	
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 50 mA dc; f = 20 MHz	$\lvert h_{fe} \rvert$			
2N1131, L				2.5	20	
2N1132, L				3.0	20	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; f = 1.0 MHz	C_{obo}		45	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; f= 1.0 MHz	C_{ibo}		80	pF
Pulse response		(see figure 3)	$t_{on} + t_{off}$		50	ns
Subgroups 5, 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups B and C electrical measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1.	Collector - base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO2}		1.0	µA dc
2.	Collector - base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO4}		2.0	µA dc
3.	Collector - emitter voltage (saturated)	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(SAT1)}$		1.5	V dc
4.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE4}			
	2N1131, L				20	45	
	2N1132, L				30	90	
5.	Base - emitter saturated voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.4.1)	$V_{BE(SAT)}$		1.3	V dc
6.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc	Δh_{FE4}	± 25 percent from initial value.		

1/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 3, and 5.

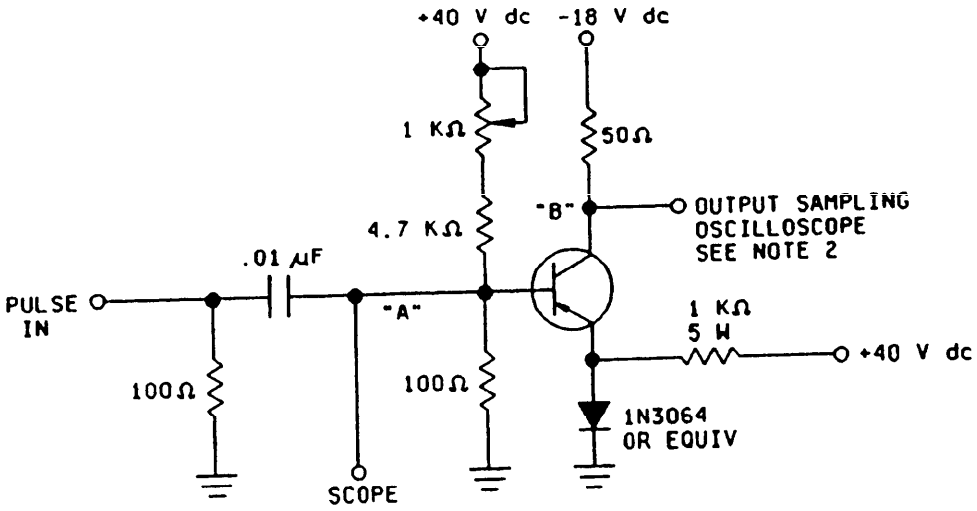
b. Subgroups 3 and 5, see table II herein, steps 2 and 4.

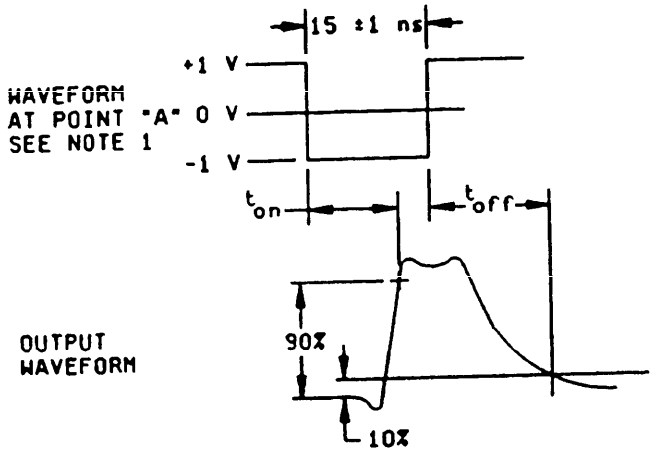
2/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroups 2 and 3, see table II herein, steps 1, 3, and 5.

b. Subgroup 6, see table II herein, steps 2 and 4.

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NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source shall be 50 Ω.
2. Sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ 12 pF, rise time ≤ .5 ns.

FIGURE 3. Nonsaturated switching time test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Terminal lead length.

d. Product assurance level and type designation.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 13, 15, 19, 85, 99

Preparing activity:
DLA - ES

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/177E	2. DOCUMENT DATE (YYMMDD) 950208

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER, TYPES 2N1131, 2N1131L, 2N1132, 2N1132L, JAN AND JANTX

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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